CYBORG SUPERMARKET ENTERPRISE DOCUMENTATION

QA Testing and Verification Report

Execution Results, Automated Test Suites, and Scenario Proofs

DOCUMENT ID: 07_QA_Testing_and_Verification_Report.pdf

Executive Publication Record

Organization:	Cyborg Omnichannel Engineering Organization
Ecosystem:	Windows POS • Android Mobile • Web Storefront • Admin Web • Backend
Primary Authority:	cyborg_system_architecture.pdf (16-Page Blueprint)
Source Dataset:	super_market(1).xlsx (1,000 Verified Egyptian Retail Records)
Security Level:	Enterprise Retail Distribution / Stakeholder Review
System Stack:	NestJS • TypeScript • PostgreSQL 16 • Redis 7 • SQLite • Flutter
Target Currency:	Egyptian Pounds (EGP) • Cairo Timezone (UTC+2 / UTC+3)

1. Quality Assurance Summary & Execution Results

The testing strategy verified 100% of all required automated test suites across backend services, POS desktop hardware logic, and end-to-end retail scenarios A through F:

Test Suite	Target Module	Total Tests	Passed	Failed	Status
Dataset Integrity	scripts/validate-dataset.js	1,000 SKUs	1,000	0	PASSED
Backend Core Services	apps/backend/src/scripts/run-b	19 Tests	19	0	PASSED
POS Hardware Logic	scripts/run-pos-tests.js	4 Tests	4	0	PASSED
E2E Scenarios (A to F)	scripts/run-e2e-scenarios.js	6 Scenarios	6	0	PASSED
Overall System Gate	scripts/run-all-tests.js	29 Verifications	29	0	PASSED (100%)

2. End-to-End Scenarios Verified

- Scenario A - Product Import: Confirmed 1,000 products imported with verified Arabic & English names.

- Scenario B - Online Shopping: Customer browsing, price recalculation, and inventory deduction verified.

- Scenario C - POS Checkout: Fast barcode scanning, 14% VAT tax, and receipt generation verified.

- Scenario D - Offline POS Sync: Local SQLite outbox queue and idempotent reconnect sync verified.

- Scenario E - Authorization: Enforcement of RBAC access restrictions verified.

- Scenario F - Data Integrity: Negative stock prevention and price tampering defenses verified.